

# Sungyeon Kook

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## Research Interests

Quantum Information Theory, Quantum Error Correction, Quantum Algorithms, Error-Correcting Codes

## Education

### Korea University

M.S. (Master of Science) in Electrical Engineering

SEOUL, KOREA

Mar. 2024 – Feb. 2026

### Korea University

B.S. (Bachelor of Science) in Electrical Engineering

SEOUL, KOREA

Mar. 2020 – Feb. 2024

## Research Experience

### Korea Institute of Science and Technology Information (KISTI)

Post-Master Researcher, Quantum Network Research Center (Advisor: Dr. Ilkwon Sohn)

DAEJEON, KOREA

Mar. 2026 – Present

### Korea University

Graduate Research Assistant, Communications and Information Systems Laboratory (Advisor: Prof. Jun Heo)

SEOUL, KOREA

Mar. 2024 – Feb. 2026

### Korea Research Institute of Standards and Science (KRISS)

Visiting Student Researcher, Center for Superconducting Quantum Computing System (Advisor: Dr. Hwan-Seop Yeo)

DAEJEON, KOREA

Aug. 2024 – Aug. 2025

### Korea University

Undergraduate Research Assistant, Communications and Information Systems Laboratory (Advisor: Prof. Jun Heo)

SEOUL, KOREA

Jan. 2023 – Feb. 2024

Undergraduate Research Assistant, Quantum Control Laboratory (Advisor: Prof. Mahn-Soo Choi)

Jul. 2023 – Aug. 2023

## Publications

- Changyeol Lee<sup>†</sup>, **Sungyeon Kook**<sup>†</sup>, Wooyeong Song, Kwangil Bae, and Ilkwon Sohn, “Efficient Quantum Modular Reduction: Crandall reduction and its Fault-tolerant resource analysis,” manuscript in preparation for submission to *Quantum Science and Technology*. (<sup>†</sup> Equal contribution.)
- Wooyeong Song, **Sungyeon Kook**, Wonhyuk Lee, and Ilkwon Sohn, “Asymmetry-aided Measurement-Based Quantum Repeaters and Distributed Quantum Computing with a Decoder-Free Client,” under review at *IEEE Network*. arXiv:2607.11238.
- Sungyeon Kook**, Yujin Kang, Ilkwon Sohn, and Jun Heo, “Low Spatial Cost CCZ Magic State Factory,” under review at *Quantum Information Processing*. arXiv:2606.24170.
- Sungyeon Kook**, Yujin Kang, and Jun Heo, “Limited Connectivity Quantum Architectures: Enhancing Surface Code Implementation Through Routing Qubits,” in *Proceedings of the 15th International Conference on Information and Communication Technology Convergence (ICTC)*, 2024. DOI: 10.1109/ICTC63764.2024.10826938.
- Yujin Kang, Youshin Chung, Huidan Zheng, **Sungyeon Kook**, and Jun Heo, “Magic State Distillation with Reduced Time Cost,” in *Proceedings of the IEEE International Conference on Quantum Computing and Engineering (QCE)*, 2024. DOI: 10.1109/QCE60285.2024.00142.

## Conferences

### Poster Presenter

- 2026 Annual Meeting of the Quantum Information Society of Korea 2026.02
- International Conference on Information and Communication Technology Convergence (ICTC) 2025.10
- Quantum Error Correction Conference (QEC 2025)** 2025.08

### Oral Presenter

- Joint Conference on Communications and Information (JCCI) 2025.04
- International Conference on Information and Communication Technology Convergence (ICTC) 2024.10
- KICS Conferences (Multiple Oral Presentations) 2023–2025

### Co-author

- Asian Quantum Information Science Conference (AQIS 2026)** 2026.08
- Optics and Photonics Congress 2026 (OPC 2026) 2026.07
- IEEE International Conference on Quantum Computing and Engineering (QCE)** 2024.09

## Research Projects

\* Journal published or under review

### Low-Spatial-Cost CCZ Magic-State Factory\*

2025–Present

- Converted the  $[[8, 3, 2]]$  CCZ distillation circuit into **eight Z-basis  $\pi/8$  Pauli-product joint measurements**, reducing the factory from **12 to 4 persistent logical patches**.
- Eliminated **eight redundant  $|0\rangle$  ancilla qubits** while preserving logical functionality and single-fault detection.
- Designed a surface-code CCZ factory using **four-qubit twisted ZY measurements**, reducing spatial cost by **71.4%** and space-time volume by **37.7%**.

### Surface-Code Decoder and Code Switching\*

2024–Present

- Built a **Stim–PyMatching decoding pipeline** for rotated surface codes, evaluating logical failure rates across code distances and physical-error rates.
- Developed an **erasure-aware MWPM decoder** by dynamically reweighting matching-graph edges using known erasure locations.
- Proposed a fault-tolerant **Bacon–Shor-to-surface-code switching protocol** for heterogeneous quantum computation.

### Architecture-Aware Quantum Circuit Mapping and Scheduling

2024–2025

- Designed heavy-hex surface-code mapping strategies to minimize SWAP overhead while balancing flag-qubit and hardware resources.
- Developed a **QODG-based scheduling framework** for checkerboard architectures by modeling gate dependencies, resource conflicts, and conditional execution latency.
- Implemented a Greedy DAG Qiskit transpiler pass, reducing average circuit depth from **73 to 61.4**.

### Fault-Tolerant Resource Analysis for Quantum Modular Reduction\*

2025–Present

- Developed a **surface-code resource-estimation framework** incorporating code distance, logical-error requirements, magic-state factories, decoder throughput, and backlog.
- Formulated a **fault-tolerant runtime model** incorporating T-depth, feed-forward latency, and decoder performance.
- Reduced estimated execution time from **53.77 ms to 30.05 ms** under Sparse Blossom decoding.

### Experimental Superconducting Quantum Computing

KRISS, 2024–2025

- Developed Python-based experimental-control software for superconducting-qubit experiments using Zurich Instruments hardware.
- Performed **qubit calibration and characterization**, including spectroscopy, randomized benchmarking, DRAG calibration, and cross-resonance gates.
- Investigated heavy-hex architectures qubit routing, flag-qubit syndrome extraction, and connectivity-aware fault-tolerant circuits.

### Variational Quantum Algorithms

2023

- Implemented VQE, QAOA, and quantum-kernel algorithms using Qiskit and PennyLane.

## Technical Skills

**Programming:** Python, C++, MATLAB

**Quantum Software:** Qiskit, PennyLane, Stim, PyMatching

**Experimental Platforms:** Zurich Instruments, Superconducting Quantum Computing

## Professional Activities

**IBM Quantum**

Sep. 2025 – Present

Qiskit Advocate

**Quantum Information Club at Korea University (QUICK)**

Jan. 2023 – Dec. 2023

Founder and Treasurer

- Co-founded the first undergraduate quantum information club at Korea University.
- Managed club administration and event organization.
- Organized the first IBM Quantum Fall Fest hosted at Korea University.

## Teaching & Honors

**Outstanding Academic Achievement**, Korea University

Spring 2023

**Teaching Assistant**, Data Science and Artificial Intelligence

Fall 2024

Provided programming and data analysis instruction for undergraduate students.

**Teaching Assistant Scholarship**, Korea University

Fall 2024